6.4.4 ASK off-chain data

Each airline may have many enterprise databases of its own, protected by firewalls. The airlines keep these databases as safe as possible and use them as appropriate for their routine operations. The airlines periodically update a display board showing available seats. The airport or the ASK consortium maintains this display, which is not a centralized database but an integrated view of the seats available on flights operated by the consortium's member airlines. In the Dapp codebase provided, this off-chain data on available seats is stored in a simple JSON file. This file format (instead of a database) is used for simplicity and to preserve this book's focus on the blockchain aspects of Dapp development. If you prefer, you can always create a MySQL or NoSQL database to store the off-chain data. The data schema, along with some sample data, is shown in figure 6.14.

FlightID	Airline	FromCity	ToCity	DepTime	SeatsAvail
1	AirlineA	BUF	NYC	6.00 AM	8
2	AirlineA	BUF	NYC	10.00 AM	6
3	AirlineA	BUF	NYC	6.00 PM	7
4	AirlineB	BUF	NYC	7.00 AM	10
5	AirlineB	BUF	NYC	1.00 PM	4
6	AirlineB	BUF	NYC	5.00 AM	2

Figure 6.14 ASK's seat availability off-chain data

6.4.5 ASK Dapp development process

You've already completed significant parts of the end-to-end development of the ASK Dapp. Figure 6.15 provides an overview of the standard development environment, the steps involved,